

Data Science for Innovation

W3: Hypothesis testing &
evaluation

Presented by
Dr Ali Anaissi

Overview of Week 3

Today: Hypothesis testing and evaluation

Objective

Overview of experimental design and learn Python tools for hypothesis testing and classifier evaluation.

Lecture

- Questions: Do two populations differ? Which approach is better?
- Significance: pairwise t-tests, non-normal data
- Evaluation: classifier evaluation

Readings

- Data Science from Scratch, Ch. 7
- Hypothesis testing (scipy lectures)
<http://goo.gl/H Cf3nP>
- Model evaluation (sklearn doco)
<http://goo.gl/Avj51Y>

Exercises

- scipy: statistical tests
- sklearn: evaluation metrics

TODO in W3

- Summarise and prepare data

Imagine..

Bob is developing a new diet

- Bob tries it on a sample of friends and family
- The mean weight loss for the diet is lower than methods compared
- Bob invests his savings in his diet startup

BUT the result was not reliable

- Followers of the diet report actually gaining weight!

What was Bob's mistake?

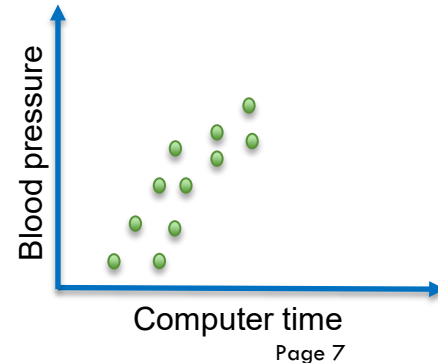
Types of Statistical Studies

Types of Statistical Studies

- Observational Study:
 - Simply observing what happens
 - Records information about subjects without applying any treatments to subjects (passive participation of researcher)
- Experimental Study:
 - Records information about subjects while applying treatments to subjects and controlling study conditions to some degree (active participation of researcher)

Observational studies

- Sample survey
 - provide information about a population based on a sample at a specific point in time. e.g.
 - Study 1: Tanning and Skin Cancer
 - The observational study involved 1,500 people.
 - Selected a group of people who had skin cancer and another group of people who did not have skin cancer
 - Asked all participants whether they used tanning beds.
 - » Wanted to see if there was an association between tanning beds and skin cancer prevalence.
 - Study 2: Average Computer Time vs Blood Pressure
 - Enroll 100 individuals in the observational study.
 - Ask them about the average computer time they spent each day.
 - Measure their blood pressure.
 - Only establish correlation not causality



Experimental Studies

- Strong hypotheses, sample size for desired power and controlled data collection per specified protocols
- Establish causality
- e.g. randomized control trials,
 - 100 subjects.
 - Factor – Average Computer Time .
 - Treatments:
 1. Control group (computer time max 30 minutes)
 2. Treatment group (computer time of 2 hours)
 - 50 subjects randomly assigned to each treatment.
 - Response : we measure the blood pressure for each group

Setting up an experiment

Experiment design

- ***Between subjects:***
Each subject sees one and only one condition
- ***Within subjects:***
Subjects see more than one or all conditions

Within Subjects

A group of people sees the test signs.



Between Subjects

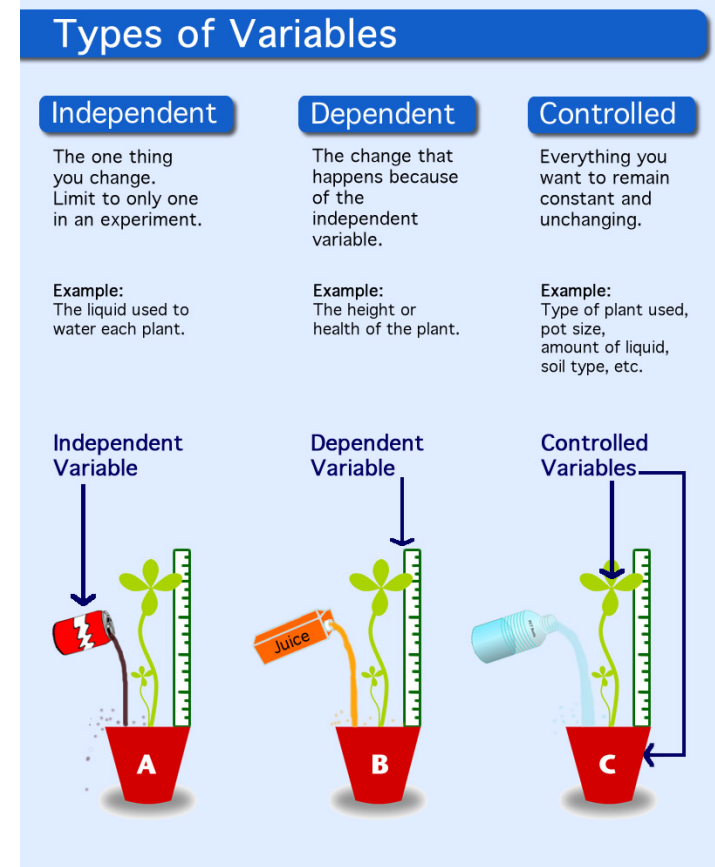
One group of people sees one set of the test signs, and a different group sees another set.



<http://www.fhwa.dot.gov/publications/research/safety/pedbike/11035/004.cfm>

Types of variables

- **Dependent variable** is the measure of interest
- **Independent variable** is manipulated to observe the effect on dependent variable
- **Controlled variables** are materials, measurements and methods that don't change



<http://edtech2.boisestate.edu/angelacovil/506/procedure.html>

Research question

- Research question (Q):
 - Asks whether the independent variable has an effect
 - “If there is a change in the independent variable, will there also be a change in the dependent variable?”
- Null hypothesis (H_0):
 - The assumption that there is no effect
 - “There is no change in the dependent variable when the independent variable changes.”

Statistical significance testing

Hypothesis testing

- We use it to specify whether to accept or reject a claim about a population depending on the evidence provided by a sample of data.
- A hypothesis test examines two opposing hypotheses about a population:
 - **The null hypothesis**
 - **The alternative hypothesis**
- A statistical test is often used to determine whether the mean of a population significantly differs from a specific value or from the mean of another population.

Testing reliability with p-values

- Most tests calculate a p-value for measuring observation extremity
- Compare to significance level threshold α
 - α is the probability of (wrongly) rejecting H_0 given that it is true
 - aka Type I error rate (false positive)
 - Commonly use α of 5% or 1%

Truth

	Decision	
	Accept H_0	Reject H_0
H_0 (No difference)	Right Decision	Type I error
H_1 (Difference exists)	Type II error	Right Decision

P-value	Indicates	Reject H_0 ?
$<\alpha$	Strong evidence against the null hypothesis	Yes
$>\alpha$	Weak evidence against the null hypothesis	No
$=\alpha$	Marginal	NA

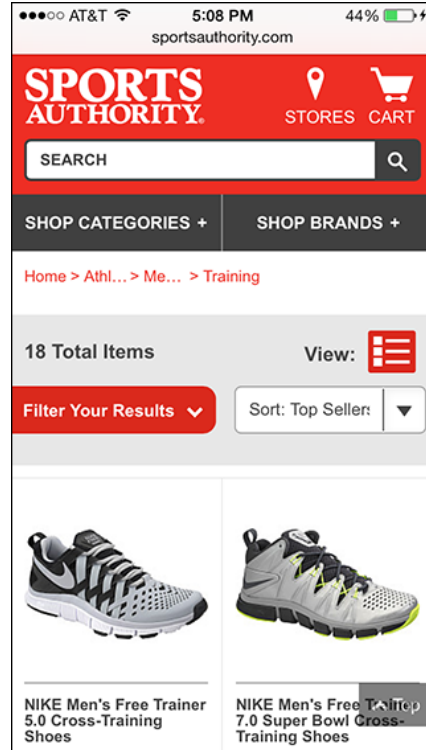
Increase the power of a significance test

- Obtain a larger sample
- Larger N means more reliable statistics
- Less likely to have errors
 - Type I: Reject true H_0
 - Type II: Fail to reject false H_0

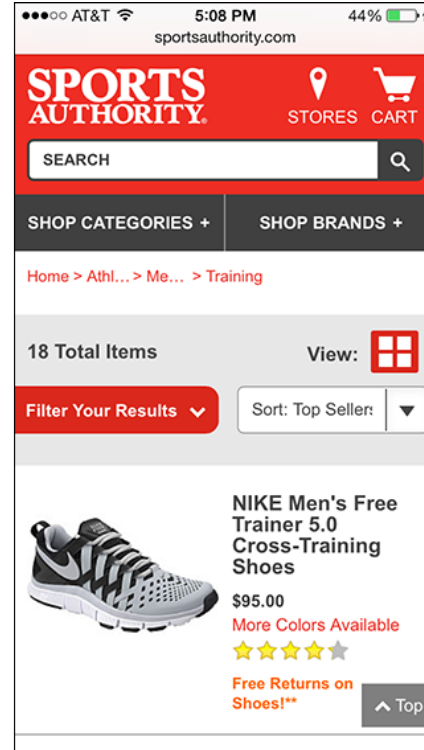
Testing which approach is better between subjects

Scenario: Comparing visual layouts

Grid view



List view



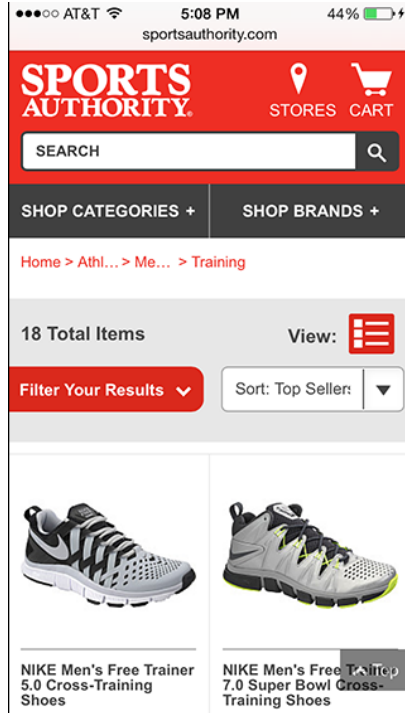
<https://www.nngroup.com/articles/image-vs-list-mobile-navigation/>

Research question

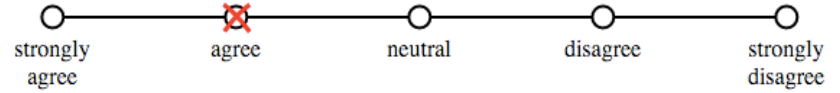
Do users prefer grid or list view?

Data/Measurement: User ratings of layouts

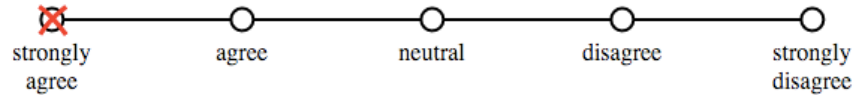
Example response
from User Group A



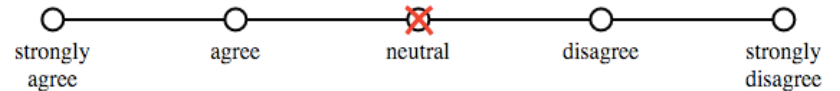
Page is easy to use.



Page gives good overview.

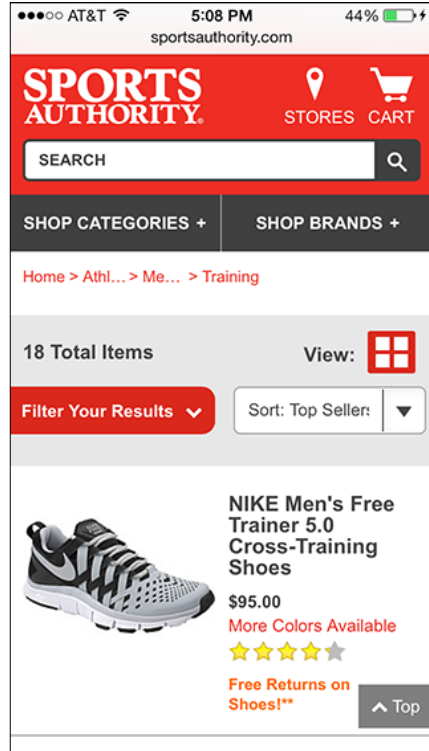


Page gives sufficient detail.

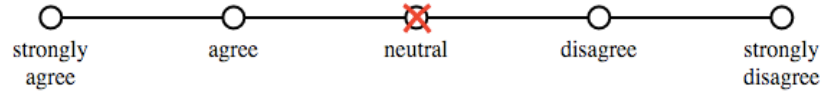


Data/Measurement: User ratings of layouts

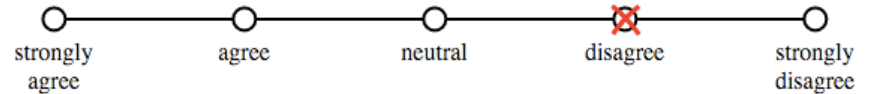
Example response
from User Group B



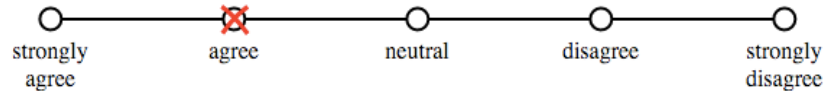
Page is easy to use.



Page gives good overview.



Page gives sufficient detail.



Setup: Comparing two versions of a display

- Subjects are users of the display (or summary, interface, etc)
 - Dependent variable is user rating (or comprehension, etc)
 - Independent variable is the version of the display
- *Problem*: Find out which version of a display is better.
- *Question*: Do users prefer Grid view?
- *Null hypothesis* H_0 : Users do not prefer Grid view.

Significance: Unpaired Student's t-test

- Tests the null hypothesis that two population means are equal
- Assumes
 - The samples are independent
 - Populations are normally distributed
 - Standard deviations are equal
- Note
 - Multiply two-tailed p-value by 0.5 for one-tailed p-value (e.g., to test $A > B$, rather than $A > B$ OR $A < B$)
- http://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.ttest_ind.html#scipy.stats.ttest_ind

Significance: Mann-Whitney U test

- Nonparametric version of unpaired t-test
- Assumes
 - The samples are independent
- Note
 - N should be at least 20
- <http://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.mannwhitneyu.html#scipy.stats.mannwhitneyu>

Exercise: Comparing layouts

- Create data
 -  code cell under “Create ratings data”
 -  code cell under “Visualise ratings data”
- Test for differences
 -  code cell under “Test whether list is preferred”
 - Do users prefer list view?
- Further exercises
 -  p-hacking example

Testing whether groups differ

Scenario: Mobile use by generation

Talking a different language					
Formative experiences	Maturists (pre-1945) Wartime rationing Rock'n'roll Nuclear families Defined gender roles - particularly for women 	Baby boomers (1945-1960) Cold War 'Swinging Sixties' Moon landings Youth culture Woodstock Family-orientated 	Generation X (1961-1980) Fall of Berlin Wall Reagan/Gorbachev/ Thatcherism Live Aid Early mobile technology Divorce rate rises 	Generation Y (1981-1995) 9/11 terrorists attacks Social media Invasion of Iraq Reality TV Google Earth 	Generation Z (Born after 1995) Economic downturn Global warming Mobile devices Cloud computing Wiki-leaks 
Attitude toward career	Jobs for life 	Organisational - careers are defined by employees	"Portfolio" careers - loyal to profession, not to employer	Digital entrepreneurs - work "with" organisations	Multitaskers - will move seamlessly between organisations and "pop-up" businesses
Signature product	Automobile 	Television 	Personal computer 	Tablet/smartphone 	Google glass, 3-D printing
Communication media	Formal letter 	Telephone 	E-mail and text message 	Text or social media 	Hand-held communication devices
Preference when making financial decisions	Face-to-face meetings	Face-to-face ideally but increasingly will go online	Online - would prefer face-to-face if time permitting	Face-to-face	Solutions will be digitally crowd-sourced

<https://ihumanmedia.com/2015/09/14/gen-x-millennials-vs-baby-boomer-real-estate-baby-work-travel-politics-shopping/>

Research question

Does mobile use differ across generations?

Data/Measurement: Survey of mobile use

- May be collected by survey or user data
- Dependent variable:
Number of texts per day
- Independent variable:
Generation {B,G,M}

Texting survey

What year were you born?

How many texts do you send per day?

Significance: Analysis of variance (ANOVA)

- Tests the null hypothesis two or more groups have the same population mean
- Assumes:
 - The samples are independent
 - Populations are normally distributed
 - Standard deviations are equal
- http://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.f_oneway.html#scipy.stats.f_oneway

Significance: Kruskal-Wallis H-test

- Nonparametric version of ANOVA
 - The test doesn't assume your data comes from a particular distribution such as normal distribution.
- Assumes samples are independent
- It is sometimes called the one-way ANOVA on ranks
 - as the ranks of the data values are used in the test rather than the actual data
- Note:
 - Not recommended for samples smaller than 5
 - Not as statistically powerful as ANOVA
 - Both ANOVA and Kruskal-Wallis H-test are extension of the Mann-Whitney test and Unpaired Student's t-test used to compare the means of more than two populations.

Setup: Comparing behavior across groups

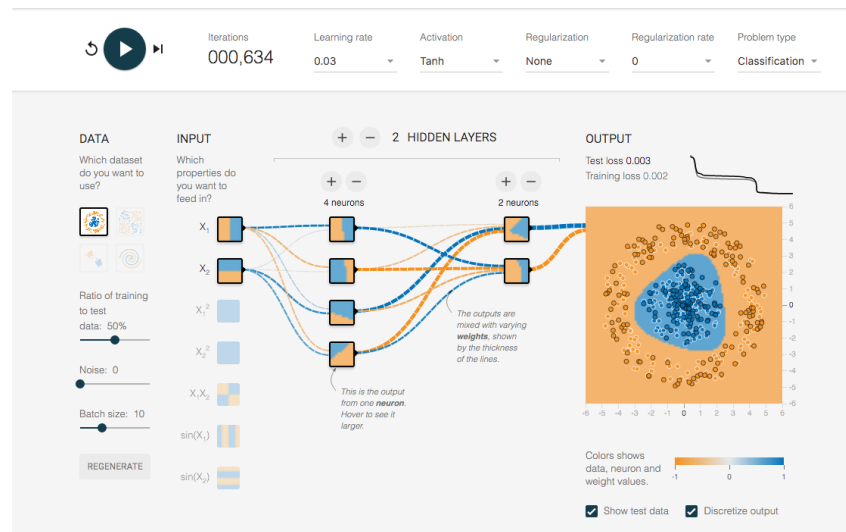
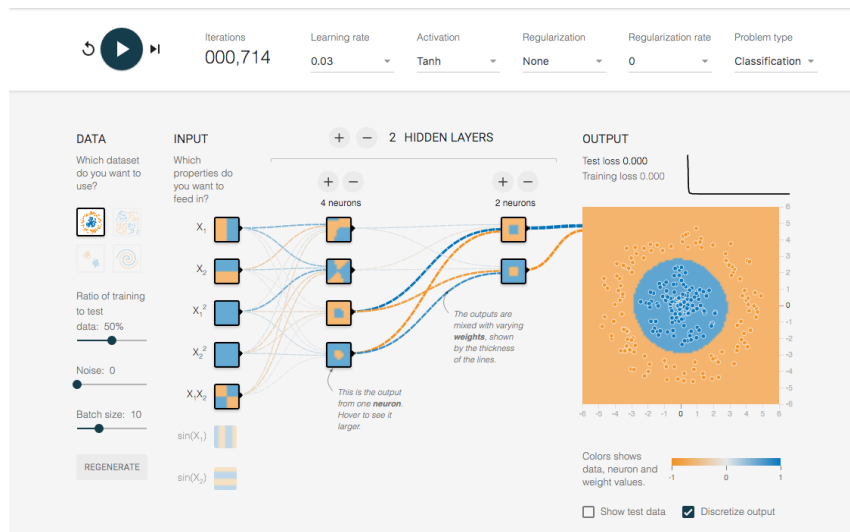
- Subjects are rows of data
 - Dependent variable is number of texts per day
 - Independent variable is generation {B,G,M}
- Q: Is there any difference between groups?
- H_0 : Group means are the same

Exercise: Comparing mobile behaviour

- Generate data
 -  code cell under “Generate generation data”
 -  code cell under “Visualise generation data”
- Test for differences
 -  code cell under “Testing for differences”
 - Does the data satisfy ANOVA assumptions?
- Further exercises
 - Do millennials and generation Z differ?

Testing which approach is better within subjects

Example scenario: Comparing classifiers



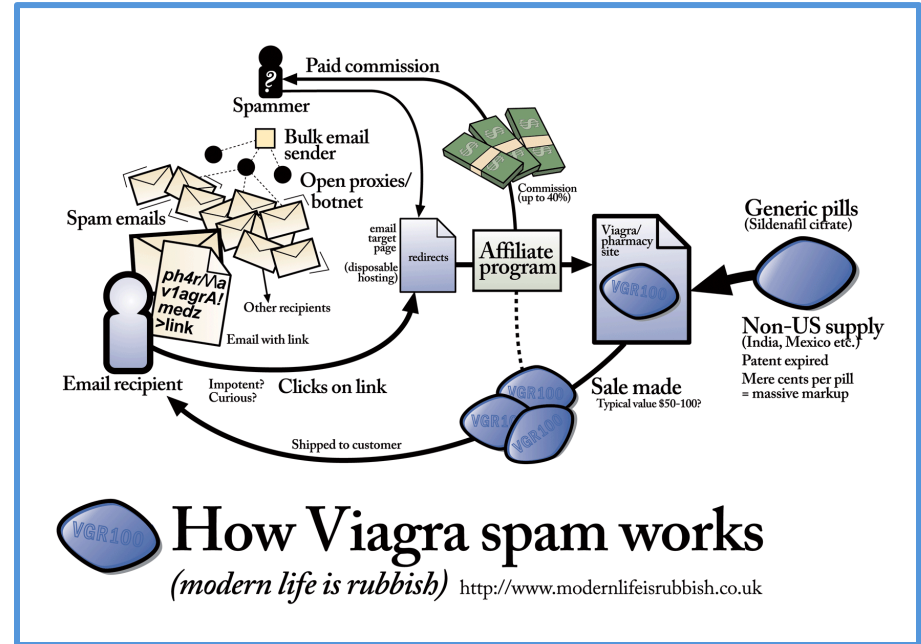
<http://playground.tensorflow.org/>

Research question

Does my new model perform better?

Task: Spam/ham detection

- Let's assume our classifiers predict whether an email is:
 - 1 (spam)
 - 0 (ham)
- Features are words, eg:
ph4rMa, v1agrA, medz



Measurement: Model evaluation

- Need to measure accuracy of system output S
- Compare to gold-standard labelling G
- Define evaluation measure: $\text{score}(S, G)$
- http://scikit-learn.org/stable/modules/model_evaluation.html#model-evaluation

Measurement: Accuracy, precision, recall, f1

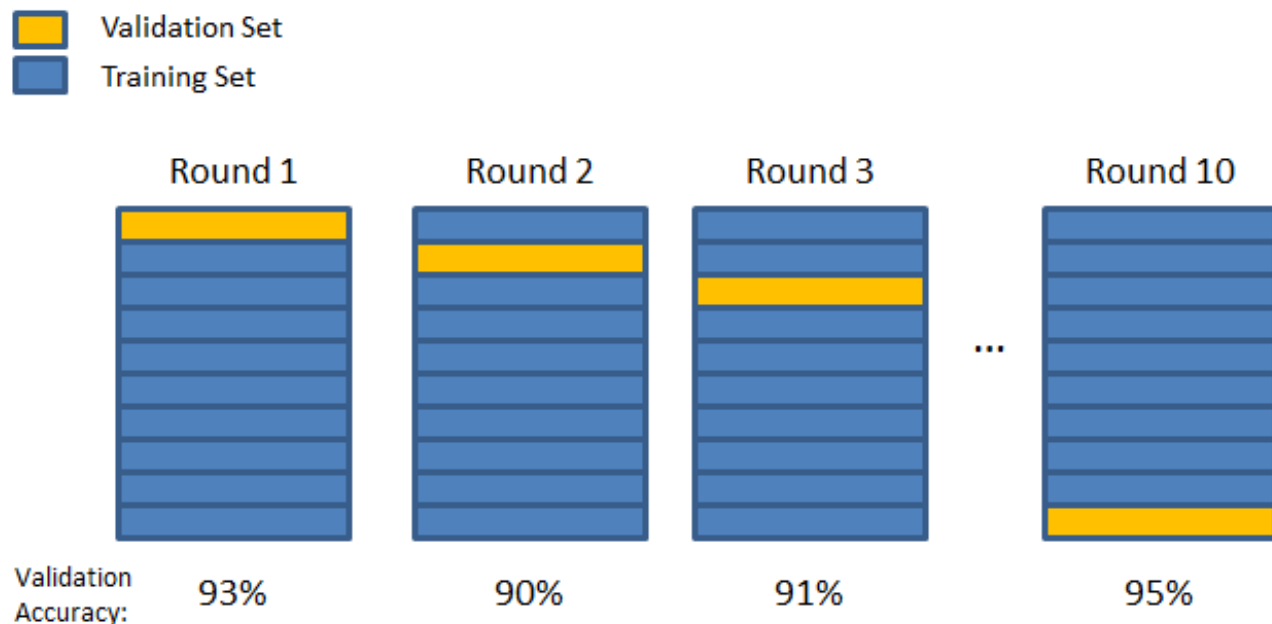
	s=1	s=0
g=1	TP (true positives)	FN (false negatives)
g=0	FP (false positives)	TN (true negatives)

- Accuracy: $(TP+TN) / N$
% correct over all instances
- Precision: $TP / (TP+FP)$
% correct system predictions
- Recall: $TP / (TP+FN)$
% correct gold labels
- F1: $2PR / (P+R)$
Harmonic mean of Precision and Recall

Evaluating Classifier Accuracy: Holdout & Cross-Validation Methods

- **Holdout method**
 - Splits the data randomly into two independent sets
 - Training set (e.g., $2/3$) for model construction
 - Test set (e.g., $1/3$) for accuracy estimation
 - Random sampling: a variation of holdout
 - Repeat holdout k times, accuracy = avg. of the accuracies obtained
- **Cross-validation** (k -fold, where $k = 10$ is most popular)
 - Randomly partition the data into k *mutually exclusive* subsets, each approximately equal size
 - Leave-One-Out is a particular form of cross-validation:
 - k folds where $k = \#$ of tuples, for small sized data

Data: Cross validation



Final Accuracy = Average(Round 1, Round 2, ...)

<https://chrisjmccormick.wordpress.com/2013/07/31/k-fold-cross-validation-with-matlab-code/>

Significance: Paired Student's t-test

- Tests the null hypothesis that two population means are equal
- Assumes
 - *The samples are paired*
 - Populations are normally distributed
 - Standard deviations are equal
- Note
 - Multiply two-tailed p-value by 0.5 for one-tailed p-value (to test $A > B$, rather than $A > B$ OR $A < B$)
- http://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.ttest_rel.html#scipy.stats.ttest_rel

Significance: Paired tests for non-parametric data

- Nonparametric version of paired t-test
- Assumes
 - The samples are paired
- Note
 - Often used for ordinal data, e.g., Likert ratings
 - N should be large, e.g., ≥ 20
- <http://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.wilcoxon.html#scipy.stats.wilcoxon>

Generate gold and classifier labellings

- We generate 10,000 gold labels
 - marking a approximately 20% as spam (1) based on a random number generator and the rest as ham (0).
 - 0: 8000 and 1: 2000
- System 1 incorrectly marks 5% of ham as spam and fails to detect 20% of actual spam:
- System 2 incorrectly marks 10% of ham as spam and fails to detect 10% of actual spam:

System 1:

		Predicted	
		1	0
Actual	1	1600	400
	0	400	7600

System 2:

		Predicted	
		1	0
Actual	1	1800	200
	0	800	7200

Setup: Comparing classifiers

- Subjects correspond to cross-validation folds
 - Dependent variable is some measure of accuracy
 - Independent variable is the algorithm, feature set, etc
- Q: Is my shiny, new model better?
- H_0 : Accuracy is not better for new model
- <http://sci2s.ugr.es/keel/pdf/algorithm/articulo/dietterich1998.pdf>

Exercise: Compare models

- Generate data
 -  code cells under “Generate gold and classifier labellings”
 -  code cell under “Split data into folds”
- Calculate accuracy
 -  code cells under “Calculating classifier accuracy”
 -  code cells under “Calculate scores across folds”
- Test for differences
 -  code cell under “Compute significance for $\text{sys1} > \text{sys2}$ ”
 - How can we manage reliability?

Review

Today: Hypothesis testing and evaluation

Objective

Learn Python tools for exploring a new data set programmatically.

Lecture

- Questions: Do two populations differ?
Which approach is better?
- Significance: pairwise t-tests, confidence intervals, non-normal data
- Evaluation: cluster evaluation, classifier evaluation, user evaluation

Readings

- Model evaluation (sklearn doco)
http://scikit-learn.org/stable/modules/model_evaluation.html#model-evaluation
- Hypothesis testing (scipy lectures)
<http://www.scipy-lectures.org/packages/statistics/index.html#hypothesis-testing-comparing-two-groups>

Exercises

- scipy: statistical tests
- sklearn: evaluation metrics

Tips and tricks

- Statistical hypothesis testing ensures results are reliable
- Experimental design includes:
 - Formulating a research question and null hypothesis
 - Designing and running experiments
 - Analysing results using appropriate statistics
- Use textbooks and documentation to find the right stats
- Sample representatively; Report p-value; Don't hack p-value
- Report precision, recall, f-score and significance

Additional reading (not examinable)

- Your favourite statistics text book (or statistician)
- Montgomery. Design and analysis of experiments.
<http://opac.library.usyd.edu.au/record=3416341>
- Robertson and Kaptein. Modern statistical methods for HCI.
<http://www.springer.com/gb/book/9783319266312>
- Hartson. The UX book. Chapters 12-18.
<http://opac.library.usyd.edu.au/record=b4415045~S4>
- Scott. Multi-armed bandit experiments.
<https://support.google.com/analytics/answer/2844870?hl=en>

Next Time

Next week: Data Mining

Objective

Learn Python tools for data mining with a focus on clustering and association rule mining.

Lecture

- Association rule mining.

Readings

- Data Science from Scratch, Ch. 11, 19

Exercises

- associations from scratch

TODO in Week 3

- Learning Python
- Explore and select project data